

A comparison of quadruple semitendinosus and patellar tendon grafts in reconstruction of the anterior cruciate ligament

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In a two-centre study, 164 patients with unilateral instability of the anterior cruciate ligament were prospectively randomised to arthroscopic reconstruction with either a patellar tendon graft using interference screw fixation or a quadruple semitendinosus graft using an endobutton fixation technique. The same postoperative rehabilitation protocol was used for all patients and follow-up at a median of 31 months (24 to 59) was carried out by independent observers. Four patients (2%) were lost to follow-up.

No significant differences were found between the groups regarding the Stryker laxity test, one-leg hop test, Tegner activity level, Lysholm score, patellofemoral pain score, International Knee Documentation Committee (IKDC) score or visual analogue scale, reflecting patient satisfaction and knee function. Slightly decreased extension, compared with the non-operated side, was found in the patellar tendon group ($p < 0.05$). Patients with associated meniscal injuries had lower IKDC, visual analogue ($p < 0.01$) and Lysholm scores ($p < 0.05$) than those without such injuries. Patients in whom reconstruction had been carried out less than five months after the injury had better final IKDC scores than the more chronic cases ($p < 0.05$).

We conclude that patellar tendon and quadruple semitendinous tendon grafts have similar outcomes in the medium term. Associated meniscal pathology significantly affects the final outcome and early reconstruction seems to be beneficial.

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The bone-patellar tendon-bone graft has been regarded as the procedure of choice in reconstruction of the anterior cruciate ligament (ACL) for more than two decades.¹⁻⁴ Alternatives include semitendinosus or combined semitendinosus/gracilis tendon grafts. Few of the numerous studies reflecting the functional outcome after these different procedures are prospective investigations and even fewer are both prospective and randomised.⁵⁻⁷ Combined semitendinosus and gracilis grafts have been used in previous randomised studies. To the best of our knowledge, no prospective, randomised study comparing bone-patellar tendon-bone grafts with quadruple semitendinosus tendon grafts has been published. The particular clinical issues relate to possible elongation of hamstring tendon grafts over time, their possible lower donor site morbidity, especially reflecting patellofemoral problems, and optimal fixation technique.

Our aim was to compare arthroscopically-assisted reconstruction of the ACL using bone-patellar tendon-bone graft and interference screw fixation with quadruple semitendinosus tendon graft using endobutton fixation proximally and screws distally.

Patients and Methods

Between 1995 and 1997, in a two-centre study, 180 patients were prospectively randomised to reconstruction of the ACL with either an ipsilateral bone-patellar tendon-bone graft or a quadruple semitendinosus graft. Inclusion criteria were single-leg insufficiency of the ACL with the trauma occurring at least two months before reconstruction, age between 15 and 45 years, no previous reconstruction of the cruciate ligaments in the involved knee and no concomitant insufficiency of the posterior cruciate ligament. We excluded 16 patients from the study because they did not meet these criteria, leaving 164 (84 with bone-patellar tendon-bone and 80 with semitendinosus tendon grafts) in the study. The patients gave their informed consent to participate and the study was approved by our ethics committee.

At reconstruction, the mean age for the 96 men and 68 women was 25.7 ± 6.9 years, the mean weight 72 ± 13 kg, and the mean height 175 ± 8 cm. The mean time from the injury to the ACL to reconstructive surgery was 16 ± 24

months (2 to 173); the median was eight months. The right knee was involved in 100 patients and the left in 64. The cause of injury was sporting activities in 148 patients (90%), other accidents in 15 and a traffic accident in one. Soccer (31%), ski-ing (23%), handball (8%) and indoor bandy (8%) were common causes. Before surgery, the patients were evaluated using the score of Lysholm and Gillquist,⁸ the activity level score of Tegner and Lysholm (present, before injury and desired level of activity)⁹ and the International Knee Documentation Committee (IKDC) scores.¹⁰ A visual analogue scale (VAS) was also used. In all clinical tests the contralateral non-injured side was used as a reference. Stryker sagittal side-to-side laxity was determined at 9.08 and 18.16 kg (OSI Stryker, Kalamazoo, Michigan).

At initial arthroscopy there were 82 patients with meniscal lesions, 47 medial and 51 lateral; 16 patients had both. Twenty-one of the meniscal injuries were sutured (13 medial and 8 lateral). Radiographs with weight-bearing views were evaluated. The ACL reconstructions were carried out by eight experienced arthroscopic knee surgeons who were all familiar with both methods.

Patellar tendon graft. In 64 patients, the central third of the patellar tendon, with 2.5 cm bone blocks, was taken through a longitudinal skin incision of approximately 5 cm. The defect in the patellar tendon was left open, but the paratenon was repaired. The cortical bone defects were not filled. In the other 20 patients, the patellar tendon graft was harvested percutaneously through one 3 cm horizontal incision at the distal pole of the patella and one 3 cm oblique incision over the tibial tuberosity. The reconstructive procedure was undertaken arthroscopically in all patients using Acufex (Acufex Microsurgical Inc, Mansfield, Massachusetts) or Arthrex (Arthrex Inc, Neaples, Florida) tibial and femoral guides for tunnel placement after the remnants of the original ligament had been removed. Notchplasty was carried out, when needed. The graft was fixed on the femoral side with a 7 × 20 mm non-absorbable interference screw. After manual tensioning of the graft and at least ten cyclic loads with full range of movement, a distal non-absorbable interference screw, 9 × 20 mm, was introduced, fixing the graft with the leg in full extension.

Semitendinosus tendon graft. The semitendinosus tendon was harvested through a 5 cm longitudinal incision over the pes anserinus with a tendon stripper (Acufex, Microsurgical Inc). The length of each tendon was between 27 and 32 cm including approximately 1 cm of periosteum from the insertion. The graft was prepared to quadruple thickness with mersilene bands in each loop (2 proximally and 1 distally), and distally with non-resorbable sutures, fixed as whip stitches in both free ends. The mean length of the graft was 7.5 cm (6.5 to 8) and the mean diameter was 9 mm (8 to 11). Fixation was by an endobutton (Acufex Microsurgical Inc) proximally and screws distally.^{11,12} We pretensioned the graft, as in the patellar tendon group, and fixed it distally with the leg in full extension. All semitendinosus

tendons were of sufficient length to make quadruple grafts and the gracilis tendon was not used concomitantly.

Postoperative management. An unlocked Genu Syncro brace (Syncro Med. GmbH, Linz, Austria) allowing a range of movement of 0° to 130° was applied for three weeks. Full weight-bearing was allowed. The same rehabilitation protocol was used for all patients, including closed-chain exercises and range of movement training from the first postoperative day. Open-chain exercises were allowed after 12 weeks. Running was allowed no earlier than 12 weeks postoperatively, and a resumption of contact sports no earlier than six months postoperatively. At a minimum of 24 months, all patients who had not sustained a complete, known rupture of the graft were asked to report to one of the assigned physiotherapists for evaluation. The same protocol as in the preoperative evaluation was used, with the addition of measurement of the difference of the circumference of the thigh 15 cm proximal to the medial joint line, and assessment by a modified patellofemoral score.¹³ The patients were also asked if they regretted their reconstruction and if they had changed occupations because of it. In addition to the one-leg-hop test, a triple-jump test was also carried out. Postoperative complications and further surgery on the reconstructed knee were recorded. In the follow-up examination both VAS (a 100 mm non-interrupted, 100 being normal and 0 being worst possible) and IKDC values (A, normal; B, nearly normal; C, abnormal; D, severely abnormal) were recorded. All scoring formulae were self-administered by the patients.

Statistical analysis. Data were summarised using descriptive statistics (mean, SD, median and range). The problem of multiplicity was discussed. Due to the exploratory nature of this study, any statistical significance was followed by a discussion of the clinical significance. No method for dealing with missing values was employed. The ordinal variables were examined by the chi-squared test. Fisher's exact test was used if the expected cell frequency was below six. The Lysholm patellofemoral and VAS scores were examined using non-parametric tests. The Wilcoxon signed-rank test was used for differences over time. The Mann-Whitney U test was used for analyses of differences between the two surgical procedures. Correlations were considered of importance if the value exceeded ($\pm$) 0.50. All tests were considered to be two-sided and a significance level of 5% was used.

Results

The two groups were similar except for the distribution of the surgical methods between the men and women, the semitendinosus tendon being used in 55 men and 25 women and the patellar tendon in 41 men and 43 women.

Of the 21 meniscal injuries which were sutured, 16 had healed at the time of reconstruction, but 36 patients (22%) presented with 38 new meniscal injuries (25 medial, 13 lateral). There were no significant differences in meniscal

injuries between the groups at the time of reconstruction. Only one patient had mild arthritic changes on preoperative radiographs (semitendinosus group).

The mean period from reconstruction to follow-up was 33 ± 7 months (24 to 59) and the median 31 months with no difference between the groups. Four patients (2%) were lost to follow-up. Five (3%) had sustained traumatic rupture of the graft in two patellar tendon grafts (1 man and 1 woman) and three semitendinosus tendon grafts (2 women and 1 man). The patellar tendon grafts ruptured two and seven months after surgery. The semitendinosus tendon graft ruptures occurred at seven, 12 and 24 months. We excluded one patient in the semitendinosus tendon graft group who had further surgery shortly before follow-up. Thus 154 patients attended for review, 80 with patellar tendon grafts and 74 with semitendinosus tendon grafts.

Complications. Two patients in the semitendinosus tendon group sustained deep infections within 14 days. Neither had positive cultures, but clinically they had septic arthritis and were successfully treated with arthroscopic lavage and antibiotics. At six months their Lysholm scores were 79 and 96. One patient in the patellar tendon group sustained a

blow-out fracture of the posterior femoral tunnel intra-operatively, but the knee was considered to be stable and follow-up radiographs showed no migration of the bone blocks. Her rehabilitation was slowed up for the first six weeks, but her Lysholm score at six months was 96.

Follow-up examination. To the question "Do you regret your reconstruction?", 143 patients (93%) answered no (70 semitendinosus and 73 patellar tendons) five patients answered yes (1 semitendinosus and 4 patellar tendons), and six patients were not sure (3 semitendinosus and 3 patellar tendons). Eight patients (5%) had changed occupations between reconstruction and follow-up because of their knee. Table I gives details of the Tegner activity level and Lysholm score before operation and at follow-up. No significant differences were found between the groups. The desired level of activity in the entire group at the time of surgery was 8 (2 to 10), but at follow-up was 7 (2 to 10), which represents a significant decrease ($p = 0.004$). VAS values are presented in Table II. No significant differences were found between the groups. The median VAS value corresponding to IKDC grade A in response to "How does your knee function?" was 98 (62 to 100),

Table I. Tegner activity level and Lysholm score preoperatively and at follow-up for patients who underwent reconstruction of the ACL either by patellar tendon or semitendinosus tendon grafts

	Patellar tendon group		Semitendinosus tendon group	
	Preop	Follow-up	Preop	Follow-up
Number of patients	84	80	80	74
Tegner activity level				
0 to 3 (% of patients)	80	23	68	24
4 to 6 (% of patients)	18	42	32	42
7 to 10 (% of patients)	2	35	0	34
Median activity level (range)	2 (1 to 10)	6 (1 to 10)	2 (0 to 6)	6 (1 to 10)
Median Lysholm score (range)	71 (32 to 90)	85 (46 to 100)	71 (29 to 90)	86 (45 to 100)

Table II. VAS scores in patients who had reconstruction of the ACL either by patellar tendon or semitendinosus tendon grafts

	Patellar tendon group		Semitendinosus tendon group	
	Preop	Follow-up	Preop	Follow-up
VAS ("How does your knee function?")				
Number of patients	46	80	53	74
Median score (range)	35 (0 to 93)	81 (0 to 100)	27 (0 to 90)	81 (0 to 100)
VAS ("How does your knee affect level of activity?")				
Number of patients	47	80	53	74
Median score (range)	20 (0 to 82)	82 (0 to 100)	20 (0 to 99)	80 (0 to 100)

Table III. Number of patients in final IKDC grades at follow-up for both groups and overall percentage

IKDC grade at follow-up	Patellar tendon (n = 80)	Semitendinosus tendon (n = 74)	% of all patients (n = 154)
A	3	4	5
B	45	37	53
C	17	19	23
D	15	14	19

Table IV. Range of movement and stability at follow-up for both groups

	Patellar tendon (n = 80)	Semitendinosus tendon (n = 74)
Extension deficit (degrees)		
<3	50	59
3 to 5	30	15
Flexion deficit (degrees)		
0 to 5	65	53
6 to 10	15	21
Lachman test (mm)		
0 (0 to 2)	40	34
+1 (3 to 5)	39	38
+2 (6 to 10)	1	2
Pivot shift test		
0	56	50
+1	23	22
+2	1	2
Stryker laxity test		
Side-to-side difference (9.08 kg)		
<-1	1	1
-1 to 2	58	45
3 to 5	21	25
>5		3
Side-to-side difference (18.16 kg)		
<-1	4	1
-1 to 2	35	31
3 to 5	38	37
>5	3	5

IKDC grade B, 83 (20 to 97), IKDC grade C, 60 (3 to 79) and IKDC grade D, 20 (0 to 36). The corresponding median VAS values in response to the question "How does your knee affect your level of activity?" was IKDC grade A, 99 (90 to 100), IKDC grade B, 87 (47 to 100), IKDC grade C, 51 (9 to 78) and IKDC grade D, 26 (0 to 52). The final IKDC evaluation is presented in Table III. The clinical findings, in comparison with the contralateral side, are shown in Table IV. Only an extension deficit of 3° or more, which was more common in the patellar tendon group, was significant ($p < 0.05$). When the anatomical zero in extension was classified as normal only three patients had 3° or more of extension loss (2 patellar and 1 semitendinosus tendon). The mean difference in the circumference of the thigh was 0.7 (1.1) cm, with no sig-

nificant difference between the groups. The one-leg-hop test revealed no significant difference between the groups. In the patellar tendon group the median value was 97% (72 to 114) of that of the contralateral leg and in the semitendinosus tendon group 96% (59 to 113). There were no differences between the groups in the triple jump test; 98% (73 to 121) and 99% (86 to 110) for the patellar tendon and semitendinosus tendon groups, respectively. No significant differences were found in the pathology of the donor site between the groups.

The patellofemoral pain scores (Table V) showed no significant differences between the groups except in the subscore 'kneeling', in which the patients with semitendinosus tendon grafts had fewer problems ($p < 0.0001$). There was no significant difference in the final IKDC, Lysholm score, Tegner activity level or patellofemoral pain score between patients operated on with open or percutaneous harvesting of the graft in the patellar tendon group. Nor were there any differences between these groups when considering the patellofemoral subscores, including ability to kneel. There was no difference in IKDC, Lysholm score or VAS values between patients with a high preinjury Tegner activity level (7 to 10) and those with low activity level before injury (0 to 6).

Additional surgery to the reconstructed knee between reconstruction and follow-up is shown in Table VI. The 30 patients who had a reconstruction less than five months after their ACL trauma had significantly better final IKDC values with more grade A and B classifications than the 124 who had waited five months or more ($p < 0.05$). The influence of associated meniscal injuries on the final IKDC and Lysholm scores (Table VII) shows significantly poorer outcomes among patients who had any type of meniscectomy at the initial arthroscopy, or at the reconstruction. No significant differences were found between men and women regarding laxity, IKDC score, Lysholm score, activity level and patellofemoral score. There were no correlations between the range of movement, laxity, compartment findings, triple jump or the one-leg-hop test and Lysholm, VAS or IKDC scores at follow-up. A correlation was found between the patellofemoral score and the Lysholm score ($r = 0.762$), as well as IKDC score ($r = 0.739$).

Table V. Median (range) modified patellofemoral pain score according to Werner et al¹³ at follow-up for both groups

	Patellar tendon graft (n = 80)	Semitendinosus tendon graft (n = 73)
Patellofemoral pain score (0, worst possible; 55, best possible)	43 (22 to 55)	43 (14 to 55)
Subscores		
Pain (0 to 5)	3 (0 to 5)	3 (0 to 5)
Pain occasional (0 to 20)	15 (0 to 20)	15 (0 to 20)
Stair-walking, up (0 to 5)	5 (2 to 5)	5 (2 to 5)
Stair-walking, down (0 to 5)	5 (4 to 5)	5 (2 to 5)
Sitting with bent knees (0 to 5)	4 (0 to 5)	4 (0 to 5)
Squatting (0 to 5)	4 (0 to 5)	4 (0 to 5)
Kneeling (0 to 5)	2 (0 to 5)	4 (0 to 5)
Catching, locking (0 to 5)	5 (0 to 5)	5 (3 to 5)

Table VI. Number of patients with or without known additional surgery between ACL reconstruction and follow-up in both groups

Type of surgery	Patellar tendon group (n = 80)	Semitendinosus tendon group (n = 80)
None	60	60
Arthroscopy with or without shaving	12	11
With notchplasty	4	1
With meniscal surgery	5	4
Removal of screw	2	2
Revision of ACL reconstruction	1	2

Table VII. IKDC and Lysholm scores of patients with and without meniscal injury

	No meniscal injury (n = 72)	Meniscal injury (n = 82)	p value
IKDC			
Final evaluation			
A	3	4	
B	48	34	
C	12	24	
D	9	20	0.006
"How does your knee function?"			
A	26	17	
B	31	32	
C	8	23	
D	7	10	0.009
"How does your knee affect your level of activity?"			
A	27	20	
B	25	21	
C	13	25	
D	7	16	0.008
Median Lysholm score (range)	89 (50 to 100)	83 (45 to 100)	0.046

Discussion

The major finding is the similarity in the final outcome between both groups. Another report,⁷ although not recording patellofemoral pain score and having differences in surgical technique, also found no significant differences in outcome scores or laxity between the patellar tendon graft and the multistranded semitendinosus graft.

Our study is unique because of the large number of patients, the high follow-up attendance frequency (98%), the prospective and randomised study design and the independent observer follow-up with self-administered questionnaires. The two-centre study design, with several experienced surgeons, also adds strength since it emphasises the comparison between the methods rather than the individual skills of surgeons. The weakness of the study is the lack of isokinetic strength tests. The higher frequency of patellar tendon grafts in women occurred because randomisation was carried out according to the method and not to both gender and method. There were, however, no differences between men and women.

The IKDC evaluation system showed that only 58% of the patients were classified as normal, or nearly normal, and 42% as abnormal or severely abnormal at follow-up. These scores are lower than those in most other studies and a possible explanation for this is our unbiased observer

follow-up in combination with self-administered questionnaires.¹⁴ Other authors have also noted a lower percentage of 'normal' or 'nearly normal' results when using the IKDC system compared with other rating systems.^{10,15,16}

Our final IKDC, VAS and Lysholm values were also comparable with a previous study of reconstruction using patellar tendon grafts.¹⁷ In our study significant improvement in knee function, activity level and IKDC and Lysholm scores was found in both groups. This was also supported by the responses to the simple question "Do you regret your operation?" to which 93% answered 'no'. It seems, however, that most patients had lowered their activity level between reconstruction and follow-up, which probably reflects the realisation that complete restoration of knee function after reconstructive surgery is very rare. Knee kinematics do not return to normal after reconstruction.¹⁸ The incidence of meniscal injuries at the time of injury to the ACL is common¹⁹ and that such injury significantly influences the outcome scores has also been demonstrated previously.^{20,21} A concomitant meniscal injury is a stronger predictive factor of outcome after reconstruction of the ACL than the choice of graft. In contrast to other reports^{6,22} the timing of surgery in our study was another factor which influenced the final outcome, with benefit if the reconstruction was undertaken within four months of injury. There may have been some selection bias, since athletes with a

higher activity level tend to undergo reconstruction earlier than recreational athletes. By contrast, no difference in outcome was noted between patients with preinjury Tegner activity levels of 7 to 10 and 0 to 6. We also found no significant difference between the grafts regarding laxity.⁵⁻⁷ In the patellar tendon group, there were a few more cases of increased stiffness compared with the non-operated side using the Stryker test (18.16 kg). This may possibly be due to the greater stiffness in the patellar tendon graft than in the semitendinosus graft.²³ The angle of the knee at which the hamstring graft should be fixed has been addressed previously.²⁴ A comparison of 30° knee flexion with 90° of knee flexion during multistranded semitendinosus graft fixation²⁵ showed that there was increased residual laxity with the smaller angle. Postoperative management and/or graft tension or avoidance of early aggressive rehabilitation with hamstring grafts was considered to be the reason for this difference. We found, however, no significant difference in side-to-side laxity despite both the semitendinosus tendon and the patellar tendon grafts being fixed with the leg in full extension. We also advocated immediate weight-bearing with a range of movement of 0° to 130°. Our clinical findings support the observation that there is no difference in strain or stress between patellar tendon and four-stranded hamstring grafts in response to cyclic submaximal loading.²⁶ The importance of minimising the distance between the fixation points to reduce creep when using different grafts has been widely discussed,^{24,27-30} but no clinical comparisons have been reported between different hamstring fixation techniques. Therefore, it is still not clear if the hamstring graft would provide less laxity if the fixation distance was minimised.

The restoration of a full range of movement after reconstruction of the ACL and its correlation with patellofemoral problems has previously been emphasised.^{20,31,32} In our study only three patients showed a lack of extension from anatomical zero and there was no significant difference in the flexion deficit between the types of graft. Although the semitendinosus tendon group was superior to the patellar tendon group in preserving hyperextension, there was no difference in the patellofemoral pain score between the groups, a result identical to previous findings.⁵ Thus, it seems that hyperextension exceeding anatomical zero is of less importance in preventing patellofemoral problems. Speculation about a lower incidence of patellofemoral problems after reconstruction with a hamstring graft compared with a patellar tendon graft^{25,33-35} has not been verified by ourselves or others.⁵⁻⁷ There is no evidence to support the suggestion of a lower incidence of persisting patellofemoral pain at a two-year follow-up using hamstring tendon grafts, compared with patellar tendon grafts. There were no differences in patellofemoral crepitus or donor site morbidity between the groups in the present study when considering the IKDC subscores, in both men and women. The correlation between hop tests and thigh muscle strength, both concentrically and eccentrically,

ly,^{19,36-38} is considered to reflect neuromuscular co-ordination.³⁹ Functional testing with one-leg-hop and triple-jump tests in our patients showed good results and also revealed no differences between the two grafts.^{17,27,38}

There were no obvious differences in the number of additional surgical procedures between reconstruction and follow-up between the groups. There were more notch-plasty procedures in the patellar tendon group, which may reflect the slightly increased incidence of extension deficits in this group.

In recent years, subjective overall rating scales or VAS scales have been used in several studies as a complement to conventional scores (Lysholm, Tegner, IKDC).^{15,17,40,41} The VAS values in our study (80 to 82), correlated well with the earlier findings.^{17,40} Patients' subjective assessment, according to IKDC, was correlated with a global rating scale,¹⁵ similar to IKDC group A in the subjective functional assessment, the value of which was 95. For IKDC groups B, C and D the values were 91, 85 and 75, respectively. In our study, a similar VAS value was found to represent IKDC subjective functional assessment group A. Groups B, C and D corresponded, however, to lower VAS values than the above.¹⁵ This difference between the studies in grading residual problems illustrates the difficulties in comparing different studies, at least when the subjective assessment is not normal and close to 100 on the VAS or global rating scale. Perhaps a 10-point scale would be better in terms of comparing different studies and reducing observation bias.

We also found no differences between men and women in outcome scores (VAS, Lysholm, IKDC, patellofemoral score).⁴² There were no correlations between values for laxity and outcome scores, which has been reported previously.¹⁶ It seems as if a certain degree of anteroposterior laxity can be tolerated as long as there is no giving-away. We interpreted the high correlation between the patellofemoral pain score and Lysholm score as well as the IKDC score to mean that pain and discomfort, rather than instability, characterise most patients with suboptimal results.

Our study showed that the patellar tendon graft with interference screw fixation and the quadruple semitendinosus tendon graft with endobutton fixation proximally and screws distally, gave equivalent results, based on both subjective and objective assessments. The only significant differences involved the ability to kneel and normalised hyperextension; both gave better results in the semitendinosus tendon group. Concomitant meniscal lesions significantly worsened the outcome and early reconstruction was beneficial. Improvements in surgical technique, as well as rehabilitation, must be sought. Until and unless further controlled clinical studies prove otherwise, we suggest that both of these methods should be considered the procedures of choice in reconstruction of the ACL.

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